

PRIME BOOKS

Math

Year 3 · Student Book

Hundreds, tens and ones · Proof & Pattern

Mathematics

Year 3 · Student Book

Ages 7 to 8 · Lower Primary · First edition 2026

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Original content. All teaching text, worked examples, practice items and explanations are original to this edition. Numerical answers were checked in Python before publication. The school syllabus in [PDF/Input](#) was used only as a scope and sequence reference.

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British English edition. Metric units and the euro throughout. Long numbers are grouped with a thin space (1 000) and the decimal separator is a point.

What is in this book

Seventeen units for the school year, following the Prime School Year 3 mathematics sequence. Work through them in order, because each unit builds on the one before.

TERM 1

1	Numbers to 1 000	Hundreds, tens and ones
2	Addition and subtraction 1	Regrouping the ones
3	Multiplication and division 1	Tables of 2, 3, 4, 5, 8, 9 and 10
4	Tally charts and frequency tables	Collecting and organising data
5	Graphs	Pictograms, bar charts, Venn and Carroll
6	Angles and movement	Turns, compass and simple maps

TERM 2

7	Addition and subtraction 2	Regrouping tens and complements
8	Measurement, area and perimeter	Length, perimeter and area
9	3D shapes	Prisms, pyramids and solid language
10	Time 1	Analogue and digital clocks
11	Time 2	Intervals and timetables
12	Money	Euros, cents and change

TERM 3

13	Fractions of shapes	Thirds, fifths, tenths and equivalence
14	More fractions	Fractions of numbers and comparing
15	Measure: mass, capacity, temperature	Scales you can trust
16	Chance	Language of likelihood
17	Pattern and symmetry	Lines of symmetry and growing patterns

How this book works

This Year 3 book is about place value that clicks, and then everything that grows from it: calculating, data, shape, measure, fractions, chance and pattern.

Show what each digit is worth. Choose a method. Check it a different way.

© HOW GROWN-UPS CAN HELP

WORK IN SHORT, HAPPY SESSIONS

Ten minutes of full attention beats an hour of sighing.
Stop while it is still fun.

ASK "HOW DO YOU KNOW?"

More often than "what is the answer?" The thinking is what stays.

DRAW, BUILD AND TALK FIRST

Let pupils use counters or sticks before they write formal sums. Hands before symbols.

COURAGE TO SAY "I AM NOT SURE YET"

That sentence is the start of learning, not the end of it.
Praise the not-knowing.

© A NOTE FOR TEACHERS

Identity: **Proof & Pattern** (slate ink, signal blue, ivory, ochre). Cover-led flat geometric style. Page budget guideline for Year 3: about 100 to 110 pages. This edition remains on the project-manager in-development list until a PM clears readiness.

Every panel has a job

Every unit in this book is built from the same few panels. Learn them once, and every topic will feel familiar.

In this unit, you will

The goals for the unit, in plain words.

Let us start

A quick warm-up to wake up your maths brain.

Worked example

A model solution with reasoning, every step shown.

Remember!

A key fact to keep, the one thing not to forget.

Thinking time

Talk and reason with a partner. No quick answer here.

I can

Check your own progress at the end of the unit.

© HOW TO WORK

Keep a pencil and notebook beside the book. Try each question before you look for help. When you finish a unit, tick only the "I can" lines you can truly do.

A wrong answer is not the end of the story. It is a clue. Check it a different way, and see where the two methods disagree. That is where the learning lives.

Before Unit 1

You need: this book, a pencil, a ruler, squared paper sometimes, and a partner for talk tasks.

© YOUR TOOLKIT

- Place-value talk: hundreds, tens and ones.
- The habit of estimating before calculating.
- Courage to say "I am not sure yet."

© HABITS THAT HELP

- Count out loud, in tens, in fives and in twos. Rhythm helps numbers stick.
- Draw the number first. Then write the sum. The picture is the thinking.
- Check every answer a second way. Two methods that agree is understanding.

© WARM-UP

- 1 Count in tens from 0 to 100.
- 2 What is 10 more than 90?
- 3 Which is larger, 47 or 74? How do you know?

UNIT 1

1

Numbers to 1 000

Hundreds, tens and ones

© IN THIS UNIT, YOU WILL

- ✓ Count on and back in ones, tens and hundreds from any number.
- ✓ Say, read and write numbers and number words from 0 to 1 000.
- ✓ Know the value of each digit in a three-digit number.
- ✓ Compare and order numbers using $<$ and $>$.
- ✓ Estimate quantities and round to the nearest 10 or 100.

◆ LET US START

Count in tens from 100 to 200. Now count in hundreds from 100 to 900. Which counting was faster? Talk to a partner about why.

UNIT 1 • PART 1

Counting and sequences

A **sequence** is a list of numbers that follows a rule.

In 15, 20, 25, the rule is "add 5", so the next two numbers are 30 and 35.

- 1 Find the rule and write the next two numbers.
 - a. 10, 20, 30, 40, ____, ____
 - b. 100, 200, 300, ____, ____
 - c. 50, 45, 40, 35, ____, ____
- 2 Count on in fifties from 150. Say the first six numbers.
- 3 Make a sequence of your own that goes down. Give it to a partner to continue.

UNIT 1 • PART 2

Place value

In a three-digit number, each digit has a **value**.

In 326 there are 3 hundreds, 2 tens and 6 ones, because $300 + 20 + 6 = 326$.



Picture 1 Hundreds, tens and ones in 326.

🔔 REMEMBER!

The place of a digit tells you its value. The 3 in 326 is worth 300, not 3.

- 1 What is each digit worth?
 - a. 415 b. 708 c. 260
- 2 Write these as hundreds, tens and ones:
 - a. 573 b. 904
- 3 Write the number that is made of 6 hundreds, 0 tens and 9 ones.

UNIT 1 • PART 3

Comparing and ordering

- 1 Which is bigger? Explain how you know.
a. 219 or 291 b. 480 or 408
- 2 Put these in order, smallest first: 219, 192, 291.
- 3 Write a number between 350 and 400.

UNIT 1 • PART 4

Estimating and rounding

Rounding to the nearest 10: look at the ones. 0 to 4 stays; 5 to 9 rounds up.

Rounding to the nearest 100: look at the tens. 0 to 4 stays; 5 to 9 rounds up.

■ WORKED EXAMPLE

Round 346 to the nearest 10 and to the nearest 100.

Nearest 10: ones digit is 6, so round up to **350**.

Nearest 100: tens digit is 4, so stay at **300**.

- 1 Round to the nearest 10: a. 43 b. 87 c. 155
- 2 Round to the nearest 100: a. 240 b. 680 c. 751
- 3 Estimate how many books would fill a shelf that already holds about 30 books and has room for "a bit more than twice that". Give a sensible range.

→ THINKING TIME

How does knowing place value help you compare two numbers quickly?

→ I CAN

- ✓ I can count on and back in ones, tens and hundreds.
- ✓ I can say what each digit is worth.
- ✓ I can compare and order three-digit numbers.
- ✓ I can continue a sequence and explain its rule.
- ✓ I can round to the nearest 10 or 100.

UNIT 1 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Numbers to 1 000** that a partner could solve.
- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Numbers to 1 000**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✦ DID YOU KNOW?

Mathematicians almost always check their work. Checking is not a lack of skill. It is how skill becomes trustworthy.

2

Addition and subtraction 1

Regrouping the ones

© IN THIS UNIT, YOU WILL

- ✓ Add a three-digit number and a one-digit or two-digit number, regrouping ones.
- ✓ Add two three-digit numbers, regrouping ones.
- ✓ Subtract with regrouping to make enough ones.
- ✓ Estimate before you calculate.

◆ LET US START

What is $9 + 6$? How could you work it out by making 10 first?

UNIT 2 • PART 1

Addition with regrouping

When ones add to 10 or more, you **regroup** 10 ones as 1 ten.

■ WORKED EXAMPLE

Add $247 + 6$.

Ones: $7 + 6 = 13$. Write 3 ones and carry 1 ten.

Tens: $4 + 1 = 5$ tens. Hundreds: 2.

So $247 + 6 = \mathbf{253}$.

- 1 Add, regrouping ones when you need to.
a. $134 + 8$ b. $256 + 7$ c. $409 + 5$
- 2 Estimate, then add: $318 + 45$.
- 3 Beatriz collects 176 stickers. Tomás gives her 28 more. How many does she have now?

UNIT 2 • PART 2

Adding two three-digit numbers

- 1 Add: a. $235 + 147$ b. $406 + 289$
- 2 In a school book fair, Class 3A sells 128 books and Class 3B sells 175 books. How many books are sold altogether?

UNIT 2 • PART 3

Subtraction with regrouping

When there are not enough ones, regroup 1 ten as 10 ones.

■ WORKED EXAMPLE

Subtract $253 - 8$.

Ones: 3 is less than 8, so regroup. 5 tens become 4 tens, ones become 13.

$13 - 8 = 5$. Tens 4, hundreds 2. Answer: **245**.

- 1 Subtract: a. $164 - 9$ b. $320 - 7$ c. $501 - 6$
- 2 Estimate, then subtract: $482 - 57$.
- 3 A tram seat has room for 48 pupils. 29 get on. How many free seats are left?

🔄 REMEMBER!

Estimate first. After you calculate, check with the inverse operation.

→ I CAN

- ✓ I can add with regrouping of ones.
- ✓ I can subtract with regrouping of ones.
- ✓ I can estimate before I calculate.

UNIT 2 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Addition and subtraction 1** that a partner could solve.
- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Addition and subtraction 1**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✧ DID YOU KNOW?

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3

Multiplication and division 1

Tables of 2, 3, 4, 5, 8, 9 and 10

© IN THIS UNIT, YOU WILL

- ✓ Recognise multiplication by 2, 5 and 10.
- ✓ Make multiplication and division fact families.
- ✓ Build and connect the 2, 4 and 8 tables.
- ✓ Build and connect the 3, 6 and 9 tables.
- ✓ Multiply single-digit numbers by 10.

◆ LET US START

How many wheels on 4 bicycles? How did you work it out?

UNIT 3 • PART 1

Exploring multiplication and division

Multiplication is equal groups. 3×5 means 3 groups of 5.

Division shares equally or groups equally. $15 \div 3 = 5$.

Fact family for 3, 5 and 15:

$3 \times 5 = 15$, $5 \times 3 = 15$, $15 \div 3 = 5$, $15 \div 5 = 3$.

- 1 Write the fact family for 2, 8 and 16.
- 2 Multiply: a. 5×7 b. 10×6 c. 2×9
- 3 Divide: a. $20 \div 5$ b. $40 \div 10$ c. $18 \div 2$

UNIT 3 • PART 2

Connecting 2, 4 and 8

Double 2s to make 4s. Double 4s to make 8s.

■ WORKED EXAMPLE

4×6 : double 2×6 . $2 \times 6 = 12$, so $4 \times 6 = \mathbf{24}$.

8×6 : double 4×6 . $24 \times 2 = \mathbf{48}$.

- 1 Complete: a. $4 \times 5 = \underline{\quad}$ b. $8 \times 3 = \underline{\quad}$ c. $8 \times 7 = \underline{\quad}$
- 2 Count in eights from 8 to 64.

UNIT 3 • PART 3

Connecting 3, 6 and 9

Double 3s to make 6s. Triple 3s to make 9s (or $3 \times 3 = 9$).

- 1 Complete: a. $3 \times 8 = \underline{\quad}$ b. $6 \times 4 = \underline{\quad}$ c. $9 \times 5 = \underline{\quad}$
- 2 Multiply by 10: a. 7×10 b. 9×10 c. 4×10
- 3 Inês packs muffins in boxes of 6. She has 54 muffins. How many full boxes?

→ I CAN

- ✓ I can use the 2, 5 and 10 tables.
- ✓ I can write fact families.
- ✓ I can connect 2, 4 and 8, and 3, 6 and 9.
- ✓ I can multiply a single digit by 10.

UNIT 3 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Multiplication and division 1** that a partner could solve.
- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Multiplication and division 1**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✦ DID YOU KNOW?

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4

Tally charts and frequency tables

Collecting and organising data

© IN THIS UNIT, YOU WILL

- ✓ Ask statistical and non-statistical questions.
- ✓ Record data with tally marks.
- ✓ Make a frequency table.
- ✓ Describe data and discuss conclusions.

◆ LET US START

How could you find out which fruit is most popular in your class without asking everyone at once in a muddle?

UNIT 4 • PART 1

Tally charts

A **tally** is a mark for each count. Bundle marks in fives: four lines and a fifth across.

FRUIT	TALLY	TOTAL
apple		12
banana		9
orange		6

- 1 What is the most popular fruit in the table above?
- 2 How many children chose banana or orange?
- 3 Make a tally chart for colours of bags in your classroom (sample at least 12 bags).

UNIT 4 • PART 2

Frequency tables

A **frequency table** shows the totals clearly.

🔔 REMEMBER!

A statistical question is answered by collecting data. "What colour is my bag?" is not statistical for the class. "What bag colours do we have?" is.

- 1 Class 3C asked: "How do you travel to school?" Results: walk 11, car 8, bus 5, bike 4.
 - a. Draw a frequency table.
 - b. How many children were asked?
 - c. Write one conclusion in a sentence.

→ I CAN

- ✓ I can make tallies in fives.
- ✓ I can complete a frequency table.
- ✓ I can say what the data shows.

UNIT 4 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Tally charts and frequency tables** that a partner could solve.
- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Tally charts and frequency tables**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✦ DID YOU KNOW?

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UNIT 5

5

Graphs

Pictograms, bar charts, Venn and Carroll

© IN THIS UNIT, YOU WILL

- ✓ Draw and read pictograms and bar charts.
- ✓ Sort data with Venn and Carroll diagrams.
- ✓ Compare what is the same and what is different.

◆ LET US START

If one picture stands for 2 books, how would you show 10 books?

UNIT 5 • PART 1

Pictograms and bar charts

In a **pictogram**, each symbol stands for a number of items.

In a **bar chart**, the height of each bar shows the total.

- 1 In a pictogram, 1 star = 2 votes. A row shows 5 stars. How many votes?
- 2 Draw a bar chart for: cats 6, dogs 9, fish 3, birds 4. Label axes.
- 3 Write two questions another pupil could answer from your bar chart.

UNIT 5 • PART 2

Venn and Carroll diagrams

A **Venn diagram** uses overlapping circles for shared properties.

A **Carroll diagram** is a yes/no sorting grid.

- 1 Sort these numbers into a Venn diagram: even numbers / greater than 20.
Numbers: 12, 15, 22, 27, 30, 18, 21.
- 2 Make a Carroll diagram: multiples of 5 / two-digit numbers.
Numbers: 5, 10, 15, 20, 25, 7, 100.

→ THINKING TIME

When is a Venn diagram clearer than a table?

→ I CAN

- ✓ I can read and draw pictograms and bar charts.
- ✓ I can sort with Venn and Carroll diagrams.

UNIT 5 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Graphs** that a partner could solve.
- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Graphs**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✦ DID YOU KNOW?

Mathematicians almost always check their work. Checking is not a lack of skill. It is how skill becomes trustworthy.

6

Angles and movement

Turns, compass and simple maps

© IN THIS UNIT, YOU WILL

- ✓ Compare angles.
- ✓ Use north, south, east and west.
- ✓ Describe position, direction and movement.
- ✓ Make a simple grid map.

◆ LET US START

Stand up. Make a quarter turn clockwise. Where are you facing now compared with the window?

UNIT 6 • PART 1

Angles

An **angle** is a turn. A right angle is a square corner. Angles can be smaller or larger than a right angle.

- 1 Find four right angles in the classroom.
- 2 Draw an angle smaller than a right angle and one larger than a right angle. Label them.

UNIT 6 • PART 2

Direction and position

Compass points: **north, south, east, west.**

- 1 If you face north and turn a right angle clockwise, which way do you face?
- 2 Describe how to go from the school gate to the library using left, right and forward steps (invent a simple path).

UNIT 6 • PART 3

Grid maps

- 1 On a 4 by 4 grid, put a school at (1, 1) and a park at (3, 4). Describe a path from school to park that only moves along grid lines.
- 2 Make a simple map of four places in your school and invent two route questions for a partner.

→ I CAN

- ✓ I can compare angles with a right angle.
- ✓ I can use compass language.
- ✓ I can follow and make a simple grid map.

UNIT 6 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Angles and movement** that a partner could solve.
- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ **UNIT PROJECT**

Make a one-page poster for **Angles and movement**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✦ **DID YOU KNOW?**

Mathematicians almost always check their work. Checking is not a lack of skill. It is how skill becomes trustworthy.

TERM 1 REVIEW

Units 1 to 6

Mixed practice. Show working. Check each answer a different way.

Number and calculating

- 1 What is each digit worth in 508?
- 2 Put in order, smallest first: 672, 627, 762, 726.
- 3 Round 485 to the nearest 10 and nearest 100.
- 4 Calculate: a. $367 + 58$ b. $420 - 75$ c. 6×8 d. $36 \div 4$
- 5 Write the fact family for 7, 5 and 35.

Data and space

- 1 In a tally, |||| |||| | means what total?
- 2 A pictogram uses 1 sun = 2 days of sunshine. Four suns mean how many days?
- 3 Face north. Turn a right angle anti-clockwise. Which way do you face?
- 4 On a grid, move from (1, 1) to (4, 3) using only right and up steps. Write one path.
- 5 Write one statistical question you could ask your class about favourite sports.

→ I CAN (TERM 1)

- ✓ I can use place value to 1 000.
- ✓ I can add and subtract with regrouping.
- ✓ I can use early multiplication tables.
- ✓ I can collect and show simple data.
- ✓ I can talk about turns and maps.

7

Addition and subtraction 2

Regrouping tens and complements

© IN THIS UNIT, YOU WILL

- ✓ Add and subtract when tens need regrouping.
- ✓ Record calculations carefully.
- ✓ Find complements of 10, 100 and 1 000.
- ✓ Use trial and improvement for missing numbers.

◆ LET US START

What do you add to 70 to make 100? What do you add to 700 to make 1 000?

UNIT 7 • PART 1

Regrouping tens

Sometimes ones are fine, but tens go past 9. Regroup 10 tens as 1 hundred.

■ WORKED EXAMPLE

Add $274 + 58$.

Ones: $4 + 8 = 12$. Write 2, carry 1 ten.

Tens: $7 + 5 + 1 = 13$ tens. Write 3 tens, carry 1 hundred.

Hundreds: $2 + 1 = 3$. Answer: **332**.

- 1 Add: a. $186 + 47$ b. $295 + 38$ c. $467 + 159$
- 2 Subtract: a. $340 - 26$ b. $523 - 78$ c. $600 - 45$

UNIT 7 • PART 2

Complements

Complements of 100 are pairs that make 100 (often using multiples of 10).

- 1 Find complements of 100: a. 40 b. 65 c. 90
- 2 Find complements of 1 000 (multiples of 10): a. 300 b. 750 c. 820

UNIT 7 • PART 3

Missing numbers

- 1 Find the missing number: a. $240 + \underline{\quad} = 300$ b. $\underline{\quad} - 45 = 155$
- 2 Use trial and improvement: $\square + 178 = 405$. Show two trials.

→ I CAN

- ✓ I can add and subtract with regrouping of tens.
- ✓ I can find complements of 100 and 1 000.
- ✓ I can find missing numbers carefully.

UNIT 7 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Addition and subtraction 2** that a partner could solve.
- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Addition and subtraction 2**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✧ DID YOU KNOW?

Mathematicians almost always check their work. Checking is not a lack of skill. It is how skill becomes trustworthy.

8

Measurement, area and perimeter

Length, perimeter and area

© IN THIS UNIT, YOU WILL

- ✓ Estimate and measure length in cm, m and km.
- ✓ Calculate perimeter of rectangles and squares.
- ✓ Introduce area by counting squares.
- ✓ Tell regular from irregular shapes.

◆ LET US START

Which is longer: a classroom door or a pencil? How would you measure each fairly?

UNIT 8 • PART 1

Units of length

100 cm = 1 m. 1 000 m = 1 km.

- 1 Measure three objects in centimetres. Round each to the nearest centimetre.
- 2 Convert: a. 3 m = ____ cm b. 250 cm = ____ m ____ cm c. 2 km = ____ m

UNIT 8 • PART 2

Perimeter

Perimeter is the distance around a shape. Add all the side lengths.

■ WORKED EXAMPLE

A rectangle is 8 cm by 5 cm.

Perimeter = $8 + 5 + 8 + 5 = 26$ cm.

- 1 Find the perimeter: a. square side 6 cm b. rectangle 12 cm by 4 cm
- 2 A playground is a rectangle 30 m by 18 m. What is its perimeter?

UNIT 8 • PART 3

Area

Area can be counted in square centimetres for small rectangles.

- 1 On squared paper, draw a rectangle 5 cm by 3 cm. Count the squares inside.
- 2 Which has the larger area: a 4 by 4 square or a 6 by 2 rectangle? Show working.

→ I CAN

- ✓ I can measure and convert lengths.
- ✓ I can find perimeters.
- ✓ I can compare areas by counting squares.

UNIT 8 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Measurement, area and perimeter** that a partner could solve.
- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Measurement, area and perimeter**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✦ DID YOU KNOW?

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9

3D shapes

Prisms, pyramids and solid language

© IN THIS UNIT, YOU WILL

- ✓ Name and build common 3D shapes.
- ✓ Compare prisms and pyramids.
- ✓ Describe faces, edges and vertices.
- ✓ Sketch 3D shapes.

✦ LET US START

Name something in the room that is shaped like a box (cuboid) and something like a ball (sphere).

UNIT 9 • PART 1

Prisms and pyramids

A **prism** has the same face running all the way through (like a triangular prism or a cuboid).

A **pyramid** comes to a point. Its base can be a triangle, square or other polygon.

- 1 Sort: cube, square-based pyramid, cylinder, triangular prism, cone, sphere. Which are prisms? Which are pyramids? Which are neither?
- 2 Build a cube and a square-based pyramid from construction kit or plasticine. What is the same? What is different?

UNIT 9 • PART 2

Faces, edges and vertices

- 1 Complete the table for a cuboid and a triangular pyramid (tetrahedron):
faces / edges / vertices.
- 2 Sketch a cylinder and label curved surface and circular faces.

🔄 REMEMBER!

A vertex is a corner. An edge is where faces meet.

→ I CAN

- ✓ I can name common 3D shapes.
- ✓ I can compare prisms and pyramids.
- ✓ I can count faces, edges and vertices.

UNIT 9 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **3D shapes** that a partner could solve.
- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **3D shapes**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✧ DID YOU KNOW?

Mathematicians almost always check their work. Checking is not a lack of skill. It is how skill becomes trustworthy.

UNIT 10

10

Time 1

Analogue and digital clocks

© IN THIS UNIT, YOU WILL

- ✓ Read and record time on analogue clocks.
- ✓ Read and record digital time.
- ✓ Match analogue and digital times.

✦ LET US START

What time do you usually start school? Say it in words.

UNIT 10 • PART 1

Analogue clocks

The short hand is the **hour** hand. The long hand is the **minute** hand.

- 1 Draw clock faces for: a. 3 o'clock b. half past 7 c. quarter past 2 d. quarter to 5
- 2 Write the time in words: minute hand on 12, hour hand on 9.

UNIT 10 • PART 2

Digital clocks

Digital time uses hours and minutes, such as 09:30.

- 1 Write digitally: a. half past 4 in the afternoon (use 24-hour if you can) b. quarter to 9 in the morning
- 2 Match: 14:15 is quarter past 2 in the afternoon.

UNIT 10 • PART 3

Matching times

- 1 Match analogue descriptions to digital:
a. half past 10 → ____
b. 07:45 → ____ (in words)
- 2 Leonor's train is at 16:20. Her analogue watch shows the long hand on 4 and the short hand just past 4. Is she on time? Explain.

→ I CAN

- ✓ I can read analogue times.
- ✓ I can read digital times.
- ✓ I can match analogue and digital times.

UNIT 10 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Time 1** that a partner could solve.
- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Time 1**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✦ DID YOU KNOW?

Mathematicians almost always check their work. Checking is not a lack of skill. It is how skill becomes trustworthy.

11

Time 2

Intervals and timetables

© IN THIS UNIT, YOU WILL

- ✓ Choose sensible units of time.
- ✓ Work out time intervals.
- ✓ Read and use simple timetables.

◆ LET US START

Would you measure a blink in hours or in seconds? Why?

UNIT 11 • PART 1

Units and intervals

Seconds, minutes, hours, days. 60 seconds = 1 minute. 60 minutes = 1 hour.

- 1 Choose a unit: a. brushing teeth b. summer holiday c. a football match
- 2 A film starts at 15:10 and ends at 16:35. How long is the film?
- 3 A lesson starts at 09:00 and lasts 50 minutes. When does it end?

UNIT 11 • PART 2

Timetables

BUS	CASCAIS	ESTORIL	LISBOA
A	08:05	08:20	09:00
B	08:35	08:50	09:30

- 1 How long does bus A take from Cascais to Lisboa?
- 2 Matilde arrives at Estoril at 08:45. Which bus can she catch?
- 3 Make a simple timetable for three events in your school morning.

→ I CAN

- ✓ I can choose sensible time units.
- ✓ I can calculate intervals.
- ✓ I can read a timetable.

UNIT 11 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Time 2** that a partner could solve.

- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Time 2**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✧ DID YOU KNOW?

Mathematicians almost always check their work. Checking is not a lack of skill. It is how skill becomes trustworthy.

UNIT 12

12

Money

Euros, cents and change

© IN THIS UNIT, YOU WILL

- ✓ Write money using euros and cents with a decimal point.
- ✓ Add amounts of money.
- ✓ Find change by subtracting.

◆ LET US START

If a pencil costs €0.40, how much do 3 pencils cost?

UNIT 12 • PART 1

Writing money

We write euros and cents with a decimal point: €3.25 means 3 euros and 25 cents.

100 cents = €1.

- 1 Write in euros: a. 250 cents b. 5 euros and 5 cents c. 90 cents
- 2 Write as cents: a. €1.20 b. €0.07 c. €4.00

UNIT 12 • PART 2

Adding money

■ WORKED EXAMPLE

$$€2.45 + €1.30 = \mathbf{€3.75}.$$

- 1 Add: a. €1.20 + €0.55 b. €3.75 + €2.25 c. €0.80 + €0.90
- 2 Rodrigo buys a notebook for €1.99 and a pen for €0.75. What is the total?

UNIT 12 • PART 3

Finding change

- 1 A book costs €4.50. Carolina pays with €5.00. What is her change?
- 2 A snack is €2.35. Afonso pays with €5.00. What is his change?
- 3 Two postcards cost €0.60 each. Salvador pays with €2.00. Change?

→ I CAN

- ✓ I can write euros and cents correctly.
- ✓ I can add amounts of money.
- ✓ I can find change.

UNIT 12 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Money** that a partner could solve.
- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Money**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✦ DID YOU KNOW?

Mathematicians almost always check their work. Checking is not a lack of skill. It is how skill becomes trustworthy.

TERM 2 REVIEW

Units 7 to 12

Calculating and measure

- 1 Find complements: a. $100 - 35$ b. $1\,000 - 450$
- 2 Add $286 + 174$. Subtract $503 - 268$.
- 3 A rectangle is 9 cm by 4 cm. Perimeter? Area in square centimetres?
- 4 How many centimetres in 3 m? How many metres in 2 km?
- 5 Name a prism and a pyramid you can see in the classroom.

Time and money

- 1 Write digitally: quarter past 6 in the morning.
- 2 A film starts at 14:20 and lasts 1 hour 25 minutes. When does it end?
- 3 Bus leaves Cascais at 08:15 and arrives Lisboa at 09:05. How long is the journey?
- 4 Total: $\text{€}2.40 + \text{€}1.75$. Change from $\text{€}5.00$ for a $\text{€}3.20$ item.
- 5 Sketch a cube and label one face, one edge and one vertex.

→ I CAN (TERM 2)

- ✓ I can regroup tens when I calculate.
- ✓ I can measure length and find perimeter.
- ✓ I can read clocks and simple timetables.
- ✓ I can work with euros and cents.

13

Fractions of shapes

Thirds, fifths, tenths and equivalence

© IN THIS UNIT, YOU WILL

- ✓ Explore thirds, fifths and tenths of shapes.
- ✓ Show equivalent fractions with diagrams.
- ✓ Link the whole and its equal parts.

◆ LET US START

If a pizza is cut into 4 equal slices and you eat 1, what fraction is left?

UNIT 13 • PART 1

Equal parts

Fractions name equal parts of a whole: halves, thirds, fifths, tenths.

- 1 Shade $\frac{1}{3}$ of a shape divided into 3 equal parts.
- 2 Shade $\frac{3}{5}$ of a shape divided into 5 equal parts.
- 3 Which diagram shows tenths? Explain how you know the parts are equal.

UNIT 13 • PART 2

Equivalent fractions

Equivalent fractions are different names for the same amount.

■ WORKED EXAMPLE

$\frac{1}{2} = \frac{2}{4} = \frac{5}{10}$, because each shades half of the whole.

- 1 Draw a strip to show that $\frac{1}{3} = \frac{2}{6}$.
- 2 Circle the equivalents of $\frac{1}{2}$: $\frac{2}{4}$, $\frac{2}{3}$, $\frac{5}{10}$, $\frac{3}{5}$.
- 3 Explain why $\frac{2}{5}$ is not equivalent to $\frac{1}{2}$.

→ I CAN

- ✓ I can show thirds, fifths and tenths of shapes.
- ✓ I can recognise simple equivalent fractions.

UNIT 13 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Fractions of shapes** that a partner could solve.

- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Fractions of shapes**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✧ DID YOU KNOW?

Mathematicians almost always check their work. Checking is not a lack of skill. It is how skill becomes trustworthy.

14

More fractions

Fractions of numbers and comparing

© IN THIS UNIT, YOU WILL

- ✓ Find fractions of quantities.
- ✓ Order and compare fractions on a number line.
- ✓ Add and subtract fractions with the same denominator within one whole.

◆ LET US START

What is half of 12? What is one third of 12?

UNIT 14 • PART 1

Fractions of numbers

To find $\frac{1}{4}$ of a number, divide by 4. To find $\frac{3}{4}$, find $\frac{1}{4}$ then multiply by 3.

■ WORKED EXAMPLE

Find $\frac{3}{5}$ of 20.

$\frac{1}{5}$ of 20 = 4, so $\frac{3}{5}$ of 20 = $3 \times 4 = 12$.

1 Find: a. $\frac{1}{2}$ of 18 b. $\frac{1}{3}$ of 24 c. $\frac{1}{5}$ of 30 d. $\frac{1}{10}$ of 40

2 Find: a. $\frac{2}{3}$ of 15 b. $\frac{3}{4}$ of 16 c. $\frac{4}{5}$ of 25

UNIT 14 • PART 2

Ordering fractions

On a number line from 0 to 1, fractions sit in order of size.

1 Place on a 0 to 1 line: $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$.

2 Use < or >: a. $\frac{1}{5}$ ____ $\frac{1}{2}$ b. $\frac{3}{10}$ ____ $\frac{2}{5}$ c. $\frac{2}{3}$ ____ $\frac{3}{4}$

UNIT 14 • PART 3

Calculating with fractions

Same denominator: add or subtract the numerators; denominator stays the same. Stay within one whole.

1 Calculate: a. $\frac{1}{5} + \frac{2}{5}$ b. $\frac{3}{8} + \frac{2}{8}$ c. $\frac{7}{10} - \frac{3}{10}$

2 A ribbon is 1 whole. You use $\frac{3}{10}$, then $\frac{2}{10}$. How much is left?

→ I CAN

- ✓ I can find fractions of quantities.
- ✓ I can compare unit fractions and simple non-unit fractions.
- ✓ I can add and subtract fractions with the same denominator within one whole.

UNIT 14 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **More fractions** that a partner could solve.
- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **More fractions**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✦ DID YOU KNOW?

Mathematicians almost always check their work. Checking is not a lack of skill. It is how skill becomes trustworthy.

UNIT 15

15

Measure: mass, capacity, temperature

Scales you can trust

© IN THIS UNIT, YOU WILL

- ✓ Estimate and measure mass.
- ✓ Estimate and measure capacity in millilitres and litres.
- ✓ Read a thermometer scale.

✦ LET US START

Which is heavier: a full water bottle or an empty one? How could you check?

UNIT 15 • PART 1

Mass

Mass is measured in grams (g) and kilograms (kg). $1\,000\text{ g} = 1\text{ kg}$.

- 1 Estimate, then check if you can: mass of a maths book, an apple, your school bag.
- 2 Convert: a. $2\text{ kg} = \text{___ g}$ b. $500\text{ g} = \text{___ kg}$ c. $1\text{ kg } 250\text{ g} = \text{___ g}$

UNIT 15 • PART 2

Capacity

Capacity is how much a container holds. Millilitres (ml) and litres (l). $1\,000\text{ ml} = 1\text{ l}$.

- 1 Which unit: a. a teaspoon b. a bath c. a juice carton
- 2 A jug holds 1.5 l. How many millilitres is that?

UNIT 15 • PART 3

Temperature

Thermometers measure temperature in degrees Celsius ($^{\circ}\text{C}$).

- 1 Read these scales (describe): warmer, 0°C , below freezing.
- 2 Morning in Serra da Estrela shows -2°C . Afternoon shows 4°C . How many degrees did it rise?

✦ DID YOU KNOW?

Water freezes at 0°C and boils at 100°C at standard pressure. These fixed points help us trust a thermometer scale.

→ I CAN

- ✓ I can use grams and kilograms.
- ✓ I can use millilitres and litres.
- ✓ I can talk about temperatures on a scale.

UNIT 15 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Measure: mass, capacity, temperature** that a partner could solve.
- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Measure: mass, capacity, temperature**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✦ DID YOU KNOW?

Mathematicians almost always check their work. Checking is not a lack of skill. It is how skill becomes trustworthy.

UNIT 16

16

Chance

Language of likelihood

© IN THIS UNIT, YOU WILL

- ✓ Use correct chance language.
- ✓ Carry out simple chance experiments.
- ✓ Describe results fairly.

✦ LET US START

Is it certain, possible or impossible that you will grow wings today?

UNIT 16 • PART 1

Chance language

Use words carefully: **impossible**, **unlikely**, **even chance**, **likely**, **certain**.

- 1 Label each event:
 - a. Rolling a normal dice and getting a number less than 7.
 - b. Rolling a 9 on a normal dice.
 - c. Pulling a red cube from a bag of 9 red and 1 blue (without looking).
- 2 Invent one event for each chance word above.

UNIT 16 • PART 2

Experiments

- 1 Toss a fair coin 20 times. Tally heads and tails. Are the totals exactly equal? Should they be?
- 2 Spin a spinner with 3 equal sections (A, B, C) twenty times. Record results. Which section won most often? Is that surprising?

🔄 REMEMBER!

Chance language describes how likely something is, not a promise of what will happen next.

→ I CAN

- ✓ I can use chance words correctly.
- ✓ I can run a simple experiment and describe the results.

UNIT 16 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Chance** that a partner could solve.

- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Chance**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✧ DID YOU KNOW?

Mathematicians almost always check their work. Checking is not a lack of skill. It is how skill becomes trustworthy.

17

Pattern and symmetry

Lines of symmetry and growing patterns

© IN THIS UNIT, YOU WILL

- ✓ Find horizontal and vertical lines of symmetry.
- ✓ Sketch a reflection.
- ✓ Extend or shorten a pattern.
- ✓ Make patterns using symmetry.

✦ LET US START

Fold a paper heart down the middle. What do you notice about the two halves?

UNIT 17 • PART 1

Symmetry

A **line of symmetry** divides a shape into two matching reflections.

- 1 Draw a square and mark all lines of symmetry you can find.
- 2 Which capital letters (A, B, C, D, E, H, M, O, T, X) have a vertical line of symmetry?
- 3 Sketch the reflection of a triangle across a vertical line.

UNIT 17 • PART 2

Patterns

- 1 Continue the pattern for three more terms: circle, square, circle, square, ...
- 2 Number pattern: 3, 6, 12, 24, ____, ____. What is the rule?
- 3 Design a border pattern that uses reflection symmetry. Colour it carefully.

→ GO FURTHER

Create a class frieze: each pupil designs a symmetrical tile that fits beside the next.

→ I CAN

- ✓ I can find horizontal and vertical lines of symmetry.
- ✓ I can sketch a simple reflection.
- ✓ I can continue shape and number patterns.

UNIT 17 • EXTRA PRACTICE

Your turn

Work in your notebook. Show your method, then check a different way.

- 1 Write three questions of your own about **Pattern and symmetry** that a partner could solve.

- 2 Solve a partner's three questions. Mark any you disagree on and talk them through.
- 3 Explain one idea from this unit to someone at home in under one minute.

→ UNIT PROJECT

Make a one-page poster for **Pattern and symmetry**. Include: one definition, one worked example with two methods, and three practice questions with answers on the back.

✧ DID YOU KNOW?

Mathematicians almost always check their work. Checking is not a lack of skill. It is how skill becomes trustworthy.

TERM 3 REVIEW

Units 13 to 17

Fractions

- 1 Shade $\frac{2}{5}$ of a bar divided into 5 equal parts.
- 2 Find: a. $\frac{1}{3}$ of 21 b. $\frac{3}{4}$ of 24 c. $\frac{2}{5}$ of 30
- 3 Calculate: a. $\frac{2}{7} + \frac{3}{7}$ b. $\frac{9}{10} - \frac{4}{10}$
- 4 Use $<$ or $>$: $\frac{1}{5}$ ____ $\frac{1}{2}$

Measure, chance, pattern

- 1 Convert: 1 kg 300 g = ____ g. 2 l = ____ ml.
- 2 Morning temperature -1°C , afternoon 6°C . What is the rise?
- 3 Label: rolling a 7 on a normal dice (impossible / even chance / certain).
- 4 Draw a shape with exactly one vertical line of symmetry.
- 5 Continue: 5, 10, 20, 40, ____, ____.
- 6 Design a symmetrical pattern using two colours.

→ I CAN (TERM 3)

- ✓ I can work with simple fractions of shapes and numbers.
- ✓ I can use mass, capacity and temperature language.
- ✓ I can talk about chance fairly.
- ✓ I can use symmetry and continue patterns.

GLOSSARY

Words to keep

digit: one of the symbols 0 to 9 used to write numbers.

place value: what a digit is worth because of its position.

sequence: a list of numbers that follows a rule.

regroup: trade 10 ones for 1 ten, or 10 tens for 1 hundred (and the reverse).

estimate: a sensible approximate answer.

round: change a number to a nearby simpler number.

perimeter: the distance around a shape.

area: how much surface a shape covers.

fraction: an equal part of a whole.

equivalent fractions: fractions that name the same amount.

tally: marks used to count as you go, bundled in fives.

frequency: how many times something occurs.

right angle: a square-corner turn.

symmetry: an exact mirror match across a line.

chance: how likely an event is.

ANSWERS

Selected answers for checking

Teachers and families: these are selected closed answers. Open tasks need discussion.

Unit 1

1a 50, 60 · 1b 400, 500 · 1c 30, 25 4a 4 hundreds, 1 ten, 5 ones · 4b 7 hundreds, 0 tens, 8 ones · 4c 2 hundreds, 6 tens, 0 ones 6 609 7a 291 · 7b 480 8 192, 219, 291 10a 40 · 10b 90 · 10c 160 11a 200 · 11b 700 · 11c 800

Unit 2

1a 142 · 1b 263 · 1c 414 3 204 stickers 4a 382 · 4b 695 5 303 books 6a 155 · 6b 313 · 6c 495 8 19 free seats

Unit 3

2a 35 · 2b 60 · 2c 18 3a 4 · 3b 4 · 3c 9 4a 20 · 4b 24 · 4c 56 6a 24 · 6b 24 · 6c 45 7a 70 · 7b 90 · 7c 40 8 9 boxes

Unit 5

1 10 votes

Unit 7

1a 233 · 1b 333 · 1c 626 3a 60 · 3b 35 · 3c 10 4a 700 · 4b 250 · 4c 180

Unit 8

2a 300 cm · 2c 2000 m 3a 24 cm · 3b 32 cm 4 96 m 6 same area (16 square cm)

Unit 11

2 1 hour 25 minutes 3 09:50

Unit 12

1a €2.50 · 1b €5.05 · 1c €0.90 3a €1.75 · 3b €6.00 · 3c €1.70 4 €2.74 5 €0.50 · 6 €2.65 · 7 €0.80

Unit 14

1a 9 · 1b 8 · 1c 6 · 1d 4 2a 10 · 2b 12 · 2c 20 5a $\frac{3}{5}$ · 5b $\frac{5}{8}$ · 5c $\frac{4}{10}$

Unit 15

2a 2000 g · 2b 0.5 kg · 2c 1250 g 4 1500 ml 6 6 degrees

Term reviews

Term 1 Q4: 425; 345; 48; 9 Term 2 Q3: perimeter 26 cm; area 36 Term 2 Q9: total €4.15; change €1.80 Term 3 Q2: 7; 18; 12 Term 3 Q3: $\frac{5}{7}$; $\frac{5}{10}$

SOURCES AND REFERENCES

Where facts came from

- Prime School Year 3 mathematics syllabus ([PDF/Input/Y3 Math Syllabus NEW.docx](#)),

used as **scope and sequence only**. No continuous prose copied.

- Place-value and calculation teaching sequences follow established primary practice as interpreted for Prime School (original examples throughout).
- Freezing and boiling points of water (0 °C and 100 °C at standard pressure): standard physical constants, checked 2026-08-05.
- Portuguese place names (Cascais, Estoril, Lisboa, Serra da Estrela) used as local context for original word problems; not taken from third-party pupil books.
- Cover and unit art files in [IMAGES/](#) produced for the Prime Books line; logo composited from the real Prime School mark, not redrawn by a model.

FOR TEACHERS

Planning notes

Audience and year band

Year 3 (ages 7–8), Lower Primary. British English. Portuguese contexts for names, places, euros and metric measure.

How this book was planned

The unit list follows the Prime School Year 3 mathematics syllabus in [PDF/Input](#) as **scope and sequence only**. Continuous prose, examples and page designs from commercial pupil books were not copied.

Pedagogy

Each unit: opener and outcomes, warm-up, short teaching, worked example where helpful, practice, extra practice, unit project, checklist. The year spine is:

> Show what each digit is worth. Choose a method. Check it a different way.

Readiness

This title is on the project-manager **in-development** list ([readiness.json](#)). Do not treat the PDF as final shipped product until a PM clears readiness after visual page review and teacher feedback ([07-REVIEW.md](#)).

Page budget

Year 3 guideline: about 100–110 pages (DEC-034). If this build is shorter, treat that as a signal to deepen practice, not as automatic readiness.

Print (when preparing KDP)

210 × 270 mm, even page count, embedded fonts, no password (DEC-035). Cover wrap is separate under [KDP/](#).

PRIME BOOKS

Mathematics

Year 3

Show what each digit is worth. Choose a method. Check it a different way.